



Rainfall can re-shape mountains

A ground-breaking discovery has proven the effect rainfall has in reshaping mountains. <http://digit.in/nov20-17>



Long-lost tectonic plate found

Geologists have located a long-lost tectonic plate - "Ressurrection", in northern Canada. <http://digit.in/nov20-18>

ceremonies. Another theory is that the lines led to sites of fire offerings to the gods, in religious shrines called wak'as. In the 1990s, geologists looking to find sources of water in the hot, dry desert noticed that the locations of underground sources of water correlated roughly with the locations of the lines. They proposed a theory that the markings on the surface was a way to track the flow of water in underground channels. Bear in mind that this theory has not been confirmed yet, but the Nazcan people did manage to live in the desert by building underground aqueducts to channel water. One of the early theories for the lines was that they were ancient irrigation systems, which has since been discounted.

Another theory is that the lines are an ancient astronomical calendar, with the motifs on the ground representing constellations in the sky, that could be seen directly above the designs at particular times of the year. For example the glyph of the monkey is thought to represent the Great Bear. The Nazca people apparently used this calendar to keep track of agrarian cycles and seasons. Another theory is that the designs are very similar to cave paintings, which have been found in the area, as well as on pottery shards from the Nazca culture.

One of the geometrical patterns, looks like the plan of a modern airport, complete with runways. This particular design is located on the top of a plateau, a suitable location for building an actual airfield. It is



The newly uncovered cat

this design, combined with a motif known as "the astronaut", which is just a standing human like figure, that gave rise to the more out theory that ancient aliens were involved in the creation of the lines. Apart from the aliens one, the other theories are all in the running, and researchers do not really know the exact function of these ancient and mysterious lines.

HOW THE LINES WERE MADE

The lines were drawn by scratching away the topsoil to expose the lighter earth right beneath the surface. This is a negative geoglyph, as against a positive geoglyph that is made by depositing material such as mud or rocks, in patterns on the ground. In the desert, the topsoil has a reddish-brown hue, which is because of the high iron oxide content of the rocks in the area. About 10 to 15 centimeters beneath the ground, is a lighter, yellowish-grey clay earth. The light coloured clay offers a sharp contrast to the dark coloured soil on top.

Typically, the figures drawn are made using one continuous, unbroken line. It remains a mystery of exactly how the ancient people managed to etch out these lines without modern equipment such as hot air balloons or aeroplanes. There

was simply no way for the people who created the lines to actually see them, with some of the designs being apparent only from an altitude of about 1,500 feet.

Researchers do have some guesses though. One of the possible explanations is that the lines were drawn using a form of surveying, where the relative two dimensional positions of the ground is determined using geometrical equations and teams that stretched ropes across the landscape. Archaeologists have unearthed wooden stakes at some of these geoglyphs, which seems to support the theory. Some of the more basic designs have been recreated on the ground using the tools and techniques known to the Nazca people, which have disproved the theories that ancient aliens had to be involved. Bear in mind that this is the same civilisation that successfully managed to live in the desert by building a complex network of aqueducts.

There may be many more lines and motifs that are undiscovered in the area. In 2019, a project by Yamagata University in Japan and IBM used machine learning to identify 143 new geoglyphs in the area. Other geoglyphs, like the newly discovered cat, may be hidden in the sides of the hills. The area continues to be a site of active archaeological explorations, and new finds may finally allow researchers to unravel the mysteries of the Nazca lines. 



The Nazca Lines that look like a runway